

Relaxation and Odd-Even Effect in Fermi Liquids

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Introduction

- k -space
- Fermi surface

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Problem Setup

- Boltzmann equation
- Fermi Dirac distribution
- Linearized collision operator

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Odd-Even effect

1

Introduction

1.1

***k*-space**

k -space

- Analogous to momentum space
- Each point $\mathbf{k}$ in k -space corresponds to a wave $e^{i\mathbf{k}\cdot\mathbf{r}}$
- **Fourier transform** converts between k -space to real space

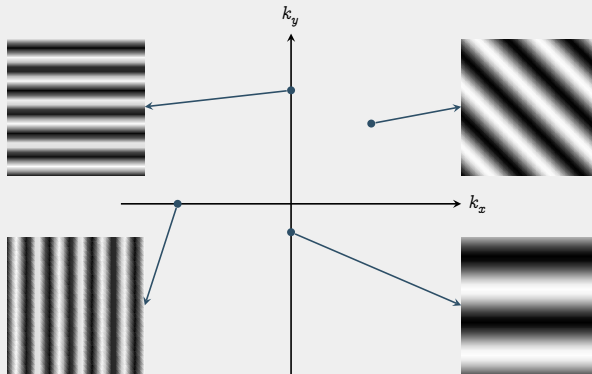


Figure 1: REPRESENTATIVE WAVES OF k -SPACE

- Distance from origin $\propto$ Wave frequency
- Direction on k -space $\implies$ Wave direction

1.2

Fermi surface

2

Problem Setup

2.1

Boltzmann equation

- Dilute gas
- Only short range interactions

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{r}} f + \mathbf{F} \cdot \nabla_{\mathbf{v}} f = \left(\frac{\partial f}{\partial t} \right)_{\text{coll}} \quad 1$$

Under the assumption of no thermal gradient, no external force, and spatial homogeneity, the Boltzmann equation reduces to

$$\frac{\partial f}{\partial t} = \left(\frac{\partial f}{\partial t} \right)_{\text{coll}} \quad 2$$

$$\left(\frac{\partial f}{\partial t}\right)_{\text{coll}} \approx \int \dots \int d^3\mathbf{p}_2 d^3\mathbf{p}'_1 d^3\mathbf{p}'_2 W(\mathbf{p}'_1, \mathbf{p}'_2 | \mathbf{p}, \mathbf{p}_2) [f_2(\mathbf{r}, \mathbf{r}, \mathbf{p}'_1, \mathbf{p}'_2) - f_2(\mathbf{r}, \mathbf{r}, \mathbf{p}_1, \mathbf{p}_2)]$$

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- $f_2(\mathbf{r}, \mathbf{r}, \mathbf{p}_1, \mathbf{p}_2)$ is the two-particle distribution function, which may be written as $f(\mathbf{p}_1)f(\mathbf{p}_2)(\mathbf{r})$
- $W(\mathbf{p}'_1, \mathbf{p}'_2 | \mathbf{p}, \mathbf{p}_2)$ is the transition rate from $(\mathbf{p}, \mathbf{p}_2)$ to $(\mathbf{p}'_1, \mathbf{p}'_2)$

In equilibrium, $\left(\frac{\partial f}{\partial t}\right)_{\text{coll}} = 0$

2.2

Fermi Dirac distribution

$$f_0(\mathbf{p}) = \frac{1}{e^{\beta(\varepsilon(\mathbf{p}) - \mu)} + 1}$$

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- $f_0(\mathbf{p})$ is the Fermi-Dirac distribution
- $\beta = 1/k_B T$ is the inverse temperature
- μ is the chemical potential

The Fermi-Dirac distribution is in equilibrium; the collision integral is zero.

2.3

Linearized collision operator

- No chemical potential

3

Odd-Even effect